EXPERIMENT 9

Deploy Web Application using Netlify and Render

AIM

To deploy a React frontend and Node.js/Express backend using Netlify and Render.

ALGORITHM

1. Push the backend and frontend projects to GitHub.
2. Deploy the backend repository on Render.
3. Set environment variables and note the backend URL.
4. Update the frontend API URL and build the React project.
5. Deploy the frontend on Netlify and test the live application.

PROGRAM

```
// Backend deployment preparation
// package.json
{
  "scripts": { "start": "node server.js" },
  "dependencies": { "express": "^4.18.0", "cors": "^2.8.5" }
}

// Start server on the host-provided port
const PORT=process.env.PORT || 5000;
app.listen(PORT,()=>console.log(`Server on ${PORT}`));

// Frontend: use deployed backend URL
const API_URL='https://your-backend.onrender.com';

// Build and upload/deploy frontend
// npm run build
// Connect GitHub repository in Netlify.
// Render: Build command = npm install
// Render: Start command = npm start
```

RESULT

Thus, the frontend and backend deployment process was completed using Netlify and Render. Verify the live URLs after deployment.

VIVA QUESTIONS

1. Why is application deployment required?
2. What is frontend hosting?
3. What is backend hosting?
4. What does npm run build do?
5. Why must CORS and production API URLs be configured?